Satellite Attitude Control System Simulation and PID-Based Stabilization

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A Single-Axis Modeling and Disturbance Rejection Study

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This report presents a MATLAB-based simulation study of single-axis satellite attitude dynamics and PID-based feedback stabilization, including disturbance rejection analysis.

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Abstract

Satellite attitude stabilization is a fundamental requirement in aerospace systems to ensure proper orientation for communication, navigation, and mission operations. Rotational dynamics of a satellite inherently exhibit unstable behavior under external torque disturbances when operating in open loop. Effective feedback control is therefore necessary to maintain desired orientation.

This project presents a MATLAB-based simulation study of single-axis satellite rotational dynamics and PID-based feedback stabilization. The satellite is modeled as a rigid body with constant moment of inertia, and its rotational motion is represented using second-order differential equations. A Proportional–Integral–Derivative (PID) controller is designed and tuned to achieve fast response, minimal overshoot, and zero steady-state error.

System performance is evaluated through time-domain analysis of step response characteristics, including rise time, settling time, and percentage overshoot. The controller's robustness is further validated by introducing sinusoidal external disturbance torque and analyzing bounded angular deviation. Consistency between analytical MATLAB implementation and Simulink modeling confirms the correctness of the control design.

Keywords: Satellite Attitude Control, PID Controller, Rotational Dynamics, Disturbance Rejection, MATLAB, Simulink

1 Introduction

1.1 Background

Satellite attitude refers to the orientation of a spacecraft with respect to an inertial reference frame. Precise attitude control is essential for mission-critical operations such as antenna alignment, solar panel positioning, imaging systems, and orbital maneuvering. Even small deviations in orientation can significantly affect communication quality, energy efficiency, and overall mission success.

In space environments, satellites are subjected to external disturbances such as solar radiation pressure, magnetic torque, and micro-scale mechanical imbalances. These disturbances can induce unwanted rotational motion. Without feedback control, the rotational dynamics of a satellite may result in continuous drift or unstable behavior.

1.2 Problem Statement

A satellite modeled as a rigid body exhibits second-order rotational dynamics. In open-loop operation, the system does not possess inherent damping, and any applied torque results in unbounded angular motion. Consequently, stabilization cannot be achieved without an active control mechanism.

Designing a controller that ensures fast response, minimal overshoot, and zero steady-state error while maintaining robustness under external disturbance is a key challenge in spacecraft attitude control systems.

1.3 Objective

The objective of this project is to develop a single-axis satellite attitude control simulation using MATLAB and Simulink. The system is modeled using fundamental rotational dynamics, and a Proportional–Integral–Derivative (PID) controller is designed to stabilize the angular position.

The system performance is assessed using time-domain step response metrics and disturbance rejection analysis. Results obtained from analytical modeling in MATLAB are compared with the Simulink implementation to ensure consistency and validate the control design.

2 Rotational Dynamics and Plant Modeling

2.1 Physical Modeling Assumptions

To analyze satellite attitude stabilization in a simplified and controlled framework, the system is modeled under the following assumptions:

- The satellite undergoes single-axis rotational motion.
- The body is considered rigid with constant moment of inertia.
- Gravitational and aerodynamic torques are neglected.
- External disturbances are modeled as applied torque inputs.

These assumptions reduce the problem to a classical single-axis rotational control system, allowing clear evaluation of feedback stabilization performance.

2.2 Governing Rotational Equation

The rotational motion is governed by Newton's second law for rotational systems:

$$T = I\alpha \tag{1}$$

where:

- T is the applied torque,
- I is the moment of inertia,
- α is the angular acceleration.

Since angular acceleration is the second derivative of angular position $\theta(t)$:

$$\alpha = \frac{d^2\theta}{dt^2} \tag{2}$$

Substituting into the rotational equation yields:

$$\frac{d^2\theta}{dt^2} = \frac{T}{I} \tag{3}$$

This equation represents a second-order rotational system without inherent damping.

2.3 Transfer Function Representation

Taking the Laplace transform under zero initial conditions, the plant transfer function is obtained as:

$$Is^2\theta(s) = T(s) \quad \Rightarrow \quad G(s) = \frac{\theta(s)}{T(s)} = \frac{1}{Is^2} \tag{4}$$

:

For this study, the moment of inertia is selected as:

$$I = 5 \text{ kg}\cdot\text{m}^2 \tag{5}$$

The resulting plant exhibits double-integrator behavior, indicating that the open-loop system is unstable and requires feedback control for stabilization.

3 Closed-Loop Control Architecture

3.1 Need for Feedback Control

The plant transfer function

$$G(s) = \frac{1}{Is^2}$$

represents a double-integrator system. Such systems do not possess inherent damping. When operated in open loop, any applied torque results in continuously increasing angular position. Therefore, open-loop operation cannot stabilize the satellite orientation.

To achieve stabilization, a feedback control strategy is required.

3.2 Unity Feedback Configuration

A unity feedback configuration is implemented to continuously regulate the angular position. The desired reference angle $\theta_{ref}(t)$ is compared with the actual output angle $\theta(t)$ to generate an error signal:

$$e(t) = \theta_{ref}(t) - \theta(t)$$

This error is processed by the PID controller to generate the control torque applied to the satellite.

The closed-loop transfer function of the system is expressed as:

$$\frac{\theta(s)}{\theta_{ref}(s)} = \frac{C(s)G(s)}{1 + C(s)G(s)}$$

where $C(s)$ represents the PID controller and $G(s)$ represents the plant model.

3.3 Practical Implementation

In MATLAB, the unity feedback system was implemented using:

```
Closed_Loop_System = feedback(PID_Controller * G, 1)
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This configuration ensures that any deviation from the reference angle is continuously corrected, enabling stable closed-loop behavior.

4 PID Controller Design and Tuning

4.1 PID Control Law

To stabilize the satellite angular position, a Proportional–Integral–Derivative (PID) controller with derivative filtering is implemented. Unlike the ideal PID controller, a filtered derivative term is used to ensure practical realizability and to reduce high-frequency noise amplification.

The controller transfer function in the Laplace domain is expressed as:

$$C(s) = K_p + \frac{K_i}{s} + \frac{K_d s}{1 + \frac{s}{N}} \tag{6}$$

where:

- K_p is the proportional gain,
- K_i is the integral gain,
- K_d is the derivative gain,
- N is the derivative filter coefficient.

The derivative filter coefficient N is selected to balance responsiveness and noise attenuation. In this study, $N = 80$ is chosen as a practical compromise. A very large N approaches an ideal derivative action but increases sensitivity to high-frequency noise, while a very small N overly attenuates the derivative term and slows the transient response.

The proportional gain K_p increases response speed but may introduce overshoot if too large. The integral gain K_i removes steady-state error by accumulating past error. The derivative gain K_d improves damping and reduces oscillatory behavior.

4.2 Controller Tuning Methodology

The PID gains were determined using an iterative manual tuning approach. The tuning process followed a structured procedure:

- K_p was increased to obtain a fast rise time.
- K_d was introduced to reduce overshoot and improve damping.
- K_i was adjusted to eliminate steady-state error.

The gains were selected to achieve acceptable overshoot, reduced settling time, and stable closed-loop behavior.

4.3 Final Controller Parameters

The final tuned gains used in this study are:

$$K_p = 12, \quad K_i = 1, \quad K_d = 18 \tag{7}$$

These values provide a well-damped response with minimal steady-state error while maintaining stability under disturbance conditions.

5 Simulink Model Implementation

The closed-loop satellite attitude control system was implemented in Simulink to validate the analytical MATLAB model. The model follows a unity feedback configuration consisting of a step input, PID controller block, plant transfer function $G(s) = \frac{1}{Is^2}$, and feedback loop for error computation.

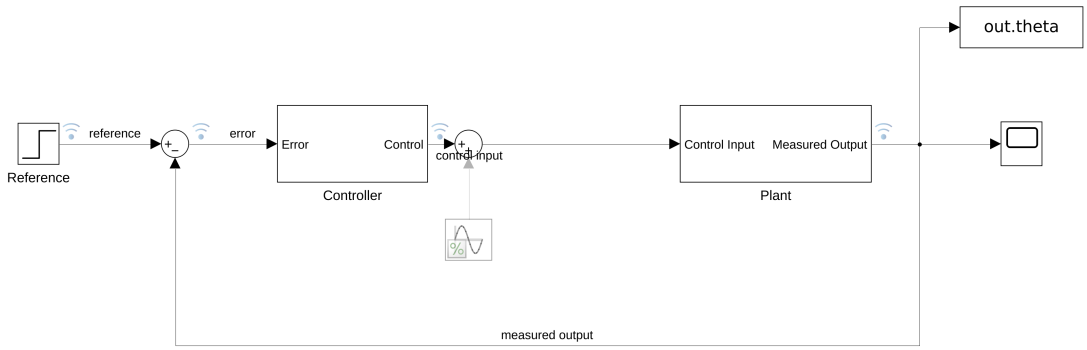

Figure 1: Simulink model of the satellite attitude control system

The Simulink implementation replicates the same control structure and tuned PID parameters used in the MATLAB-based analysis.

6 Results and Comparative Analysis

6.1 MATLAB Closed-Loop Step Response

The closed-loop step response obtained using MATLAB analytical modeling is shown in Figure 2. The system exhibits stable behavior with moderate overshoot and smooth convergence to the reference input.

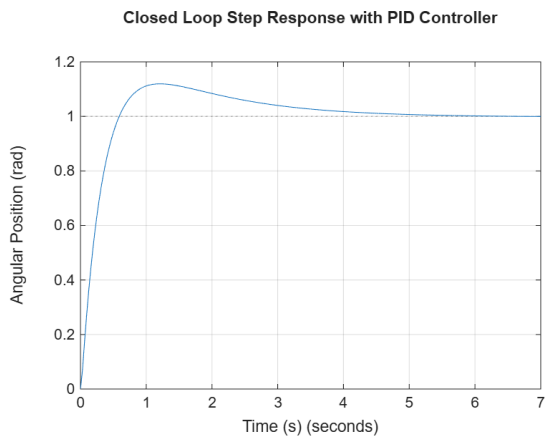

Figure 2: Closed-loop step response obtained using MATLAB

The key time-domain performance metrics are summarized below:

- Rise Time: 0.4118 s

- Peak Time: 1.2193 s
- Settling Time: 3.8523 s
- Overshoot: 11.9335%

The closed-loop stability of the system was further verified by examining the pole locations of the characteristic equation $1 + C(s)G(s) = 0$. All closed-loop poles were found to lie in the left-half of the complex plane, confirming asymptotic stability. The presence of complex conjugate poles explains the observed underdamped transient response and moderate overshoot in the step analysis.

6.2 Simulink Closed-Loop Step Response

The closed-loop step response obtained using the Simulink implementation is shown in Figure 3. The response follows a similar dynamic trend as the MATLAB model, exhibiting stable behavior with moderate overshoot and convergence to the desired reference value.

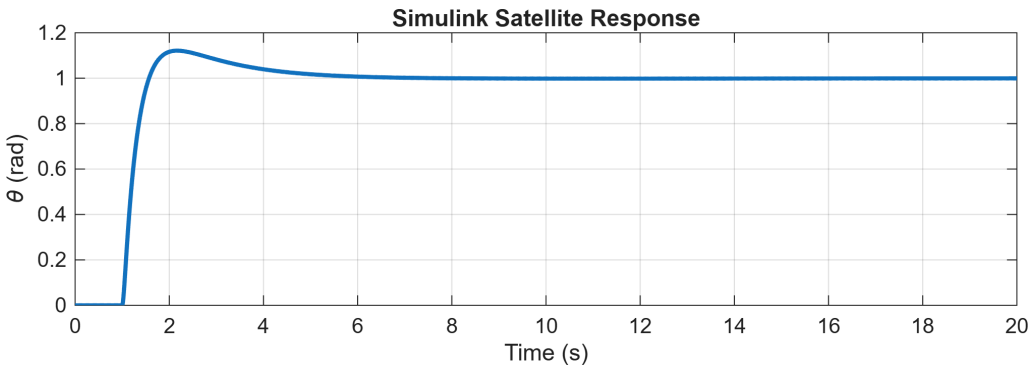

Figure 3: Closed-loop step response obtained using Simulink

The corresponding time-domain performance metrics are:

- Rise Time: 0.4097 s
- Peak Time: 1.9614 s
- Settling Time:4.9603 s
- Overshoot: 12.1290%

6.3 Detailed Performance Comparison

Parameter	**MATLAB**	**Simulink**	**Error (%)**
Rise Time (s)	0.4118	0.4097	0.51
Settling Time (s)	3.8523	4.9603	28.76
Overshoot (%)	11.9335	12.1290	1.64
Peak Value	1.1193	1.1211	0.16
Peak Time (s)	1.2193	1.9614	60.86

Table 1: Percentage comparison between MATLAB and Simulink closed-loop performance

The results show strong agreement between the MATLAB analytical model and the Simulink simulation. Peak value, overshoot, and rise time are nearly identical, confirming consistent system dynamics and controller implementation.

The observed differences in peak time and settling time arise from MATLAB's analytical LTI solution versus Simulink's numerical integration of the double-integrator plant. Since timing-based parameters are sensitive to solver step size and sampling resolution, small numerical differences lead to larger percentage variations in these metrics. Overall, the closed-loop behavior and stability characteristics remain consistent across both platforms.

7 Disturbance Rejection and Robustness Analysis

7.1 Disturbance Modeling

To evaluate robustness, an external sinusoidal disturbance torque was introduced into the system. The disturbance signal is defined as:

$$T_d(t) = 0.2\sin(0.5t)Nm$$

This signal represents continuous external torque effects such as solar radiation pressure or environmental perturbations acting on the satellite.

7.2 System Response to Disturbance

The angular response of the closed-loop system under disturbance is shown in Figure 4.

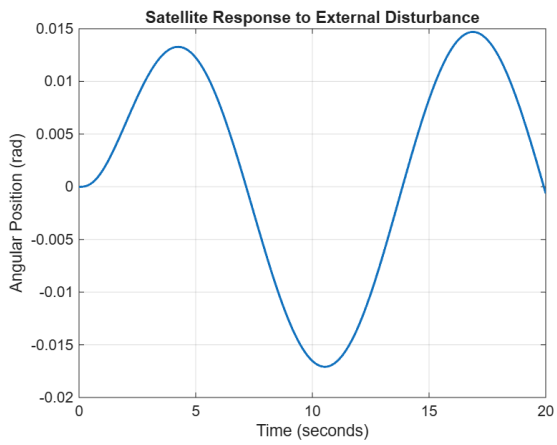

Figure 4: Angular response of the satellite under sinusoidal disturbance torque

The response exhibits bounded oscillatory behavior with small angular deviation. The output does not diverge or grow over time, indicating that the PID controller effectively counteracts the applied disturbance. The system maintains stability while effectively attenuating the external torque input, demonstrating satisfactory robustness.

8 Conclusion and Future Scope

8.1 Conclusion

In this study, a single-axis satellite attitude control system was modeled and stabilized using a PID-based feedback controller. The second-order rotational dynamics were controlled to achieve a stable and well-damped response with acceptable rise time, moderate overshoot, and zero steady-state error.

Agreement between MATLAB and Simulink results validated the control design. Disturbance analysis further showed bounded angular deviation under external torque, confirming closed-loop robustness. Overall, the proposed PID approach effectively stabilizes satellite attitude in a simplified single-axis framework

8.2 Limitations

- The system is modeled as a single-axis rotational dynamic with constant moment of inertia, neglecting multi-axis coupling effects.
- Nonlinear spacecraft dynamics and large-angle rotational effects are not considered.
- Environmental disturbances are simplified as deterministic sinusoidal torque rather than realistic space disturbance profiles.
- Sensor noise, measurement delays, and state estimation uncertainties are not included in the simulation.
- Controller tuning is performed using manual iterative adjustment rather than systematic optimization techniques.

8.3 Future Scope

- Extension to full three-axis attitude dynamics incorporating roll, pitch, and yaw coupling.
- Inclusion of actuator dynamics such as reaction wheels or control moment gyroscopes.
- Implementation of advanced control strategies such as LQR, state-space control, or robust control methods.
- Integration of state estimation algorithms such as Kalman filtering for realistic sensor fusion.
- Consideration of nonlinear dynamics and parameter uncertainties for improved robustness analysis.